



RACE PERFORMANCE ANALYTICS

Strategy Division · Performance Analytics Report

DATA SOURCE

Snowflake F1_DB

COVERAGE

1950 – 2024

TOTAL RACES

1,125 Races

Tools: Snowflake · SQL · Power BI Desktop · Power BI Service · Claude AI

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Three findings that define 75 years of Formula 1

QUALIFYING DOMINANCE

79%

of wins from top 3 grid slots

- Pole position converts to victory 42% of the time
- Top 3 grid positions account for 79% of all race wins
- Win probability drops sharply after P3

RELIABILITY REVOLUTION

89%

modern finish rate (2024)

- Finish rates risen from 42% in 1980 to 89% in 2024
- Mercedes leads at 90% finish rate, 3.1% mech DNF
- Toro Rosso least reliable at 73% finish rate

DRIVER DOMINANCE

105

Hamilton wins — modern era

- Hamilton leads with 105 wins & 202 podiums since 2000
- Two dominance eras: Ferrari 2002–04, Mercedes 2014–19
- 2008, 2010, 2012 — most contested title fights

F1 RACE PERFORMANCE ANALYTICS

Project Scope & Data Sources

Strategy Division · 1950–2024 · Snowflake: F1_DB.ANALYTICS

DATA ARCHITECTURE

Source:	Snowflake Data Warehouse
Database:	F1_DB
Raw schema:	F1_DB.RAW (14 tables)
Analytics schema:	F1_DB.ANALYTICS (5 views)
BI Tool:	Power BI Desktop + Service
AI Layer:	Claude API — insight generation

14 SOURCE TABLES

results	26,759 rows	lap_times	589,081 rows
driver_standings	34,863 rows	races	1,125 rows
pit_stops	11,371 rows	qualifying	10,494 rows
drivers	861 rows	constructors	212 rows
constructor_standings	13,391 rows	status	139 rows
circuits	77 rows	seasons	75 rows

DATA PIPELINE



F1 RACE PERFORMANCE ANALYTICS

Q1 · Grid Position vs Race Outcome

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Pole to Win %

42%

All time conversion

Front Row Win Share

66%

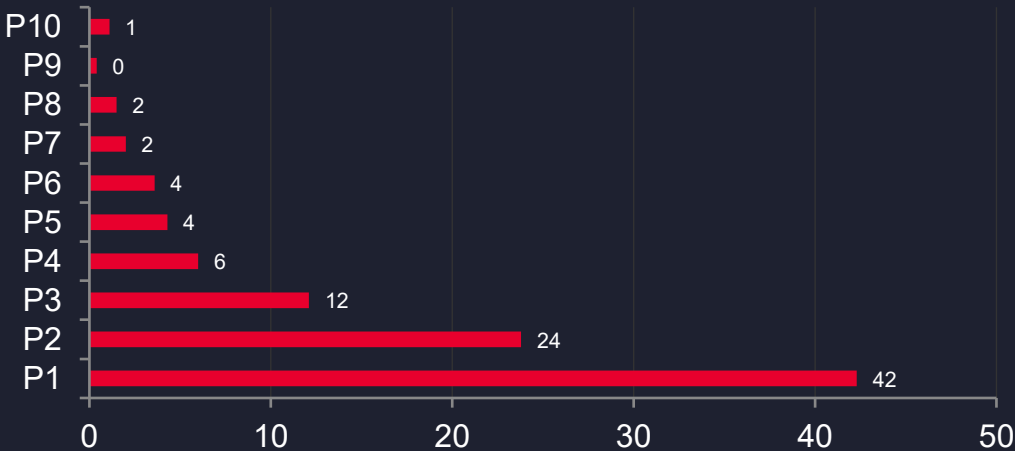
P1 + P2 combined

Top 3 Grid Win Share

79%

P1 + P2 + P3

Win conversion rate by starting grid position



KEY FINDINGS — Q1

- Pole converts to win 42% of the time — more than double any other grid slot
- Top 3 positions account for 79% of all race wins across 1,125 races
- Win probability drops sharply from 42% at P1 to just 0.4% at P9
- This relationship is consistent across all eras from 1950–2024

KEY FINDING

Qualifying is the single strongest predictor of race outcome. 79% of F1 races are won from the top 3 grid positions — teams must prioritise one-lap pace above all other performance metrics.

F1 RACE PERFORMANCE ANALYTICS

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Q2 · Constructor Reliability (2000–2024)

Modern Finish Rate

77%

2000–2024 average

Best: Mercedes

90%

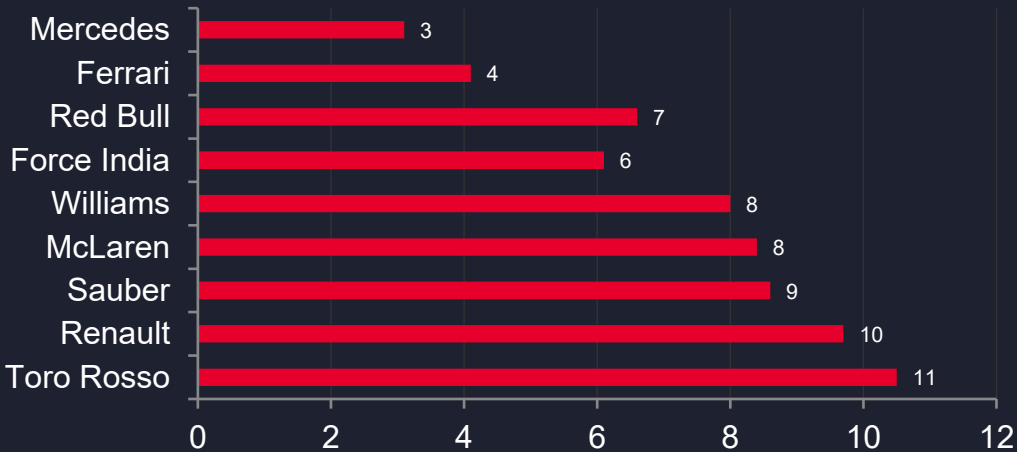
Finish rate

Total DNFs (Modern)

2,329

Since year 2000

Mechanical DNF rate by constructor (2000–2024)



KEY FINDINGS — Q2

- Mercedes leads reliability — 90% finish rate, 3.1% mech DNF rate
- Toro Rosso least reliable — 73% finish rate, 10.4% mech DNF rate
- Force India: highest accident DNF at 11.6% — double the field avg
- F1 finish rates improved: 42.6% in 1980 → 88.7% in 2024

KEY FINDING

Reliability gap between top and bottom constructors is significant. Mercedes (3.1%) vs Toro Rosso (10.4%) mechanical DNF rate translates directly to lost championship points every season.

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Q3 · Driver Performance (2000–2024)

Total Drivers (All Time)

861

Across all seasons

Hamilton Race Wins

105

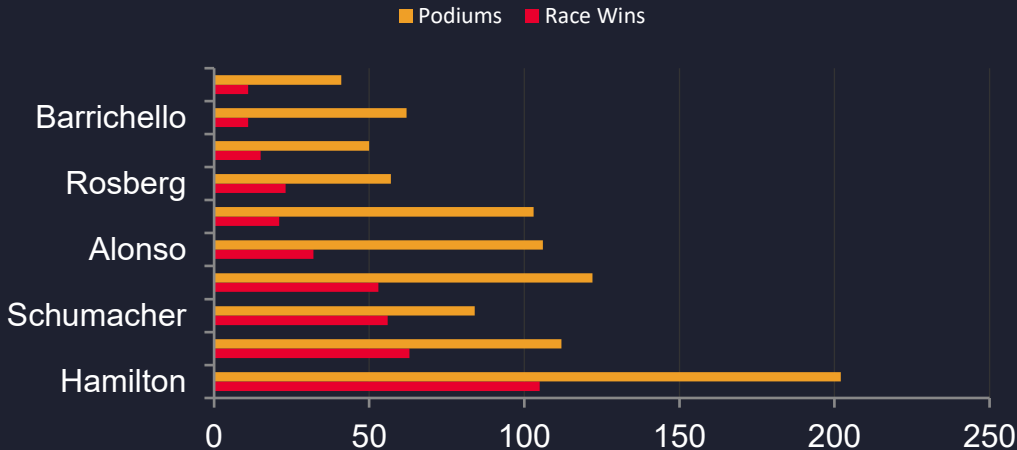
Modern era leader

Max Positions Gained

+18

Best single race comeback

Top 10 drivers — wins & podiums (2000–2024)



KEY FINDINGS — Q3

- Hamilton leads modern era — 105 wins & 202 podiums since 2000
- Verstappen shows fastest wins accumulation among current drivers
- Schumacher & Vettel show dominant multi-year championship periods
- Points per race reveals true consistency beyond raw win counts

KEY FINDING

Driver win rate relative to car performance is more meaningful than raw win counts. Alonso and Räikkönen consistently outperformed their machinery throughout their careers.

F1 RACE PERFORMANCE ANALYTICS

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Q4 · Championship Competitiveness (1980–2024)

Most Contested Season

2012

P1 vs P2 gap < 50pts

Largest Dominance Gap

2023

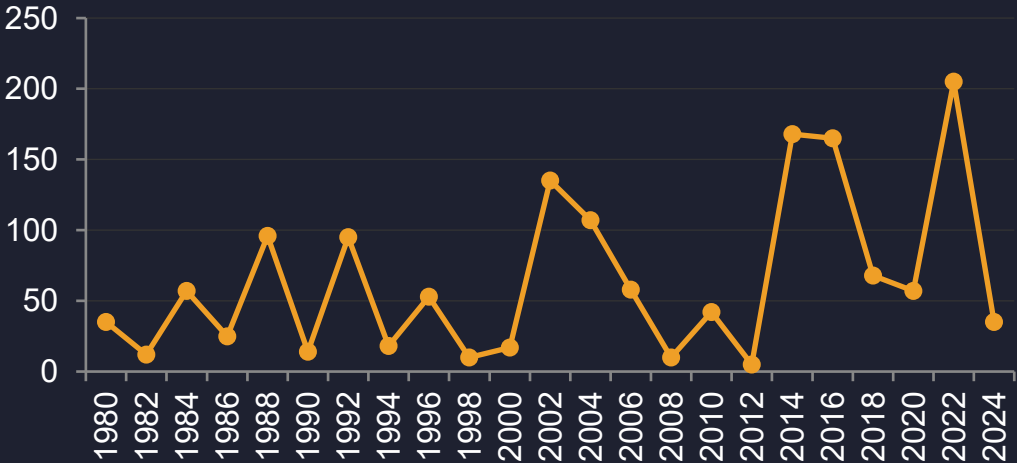
Red Bull 250pt+ margin

Dominance Periods

3

Ferrari · Mercedes · Red Bull

Constructor championship P1 vs P2 points gap by season



KEY FINDINGS — Q4

- Ferrari (2002–2004) & Mercedes (2014–2020) — two clear dominance periods
- Most contested seasons: 2008, 2010, 2012 — gaps below 50 points
- 2022–2023 Red Bull dominance approaching historic levels — 250pt+ gaps
- Championship competitiveness has declined as constructor budgets diverge

KEY FINDING

Regulatory changes drive competitive resets. The 2022 ground effect rules temporarily redistributed performance — but Red Bull adapted fastest, creating a new dominance period within 2 seasons.

F1 RACE PERFORMANCE ANALYTICS

Methodology

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How this analysis was conducted

01

Data Extraction

14 tables loaded into Snowflake F1_DB.RAW schema via CSV staging. Data covers 1950–2024 from the Ergast Motor Racing Database.

02

SQL Analytics Layer

5 analytical views built in F1_DB.ANALYTICS schema. NULLIF() used to handle \N string nulls. All joins validated against row counts.

03

Power BI Dashboard

Power BI Desktop connected directly to Snowflake. DAX measures built for KPIs. 4-page report published to Power BI Service workspace.

04

AI-Assisted Analytics

Claude API used for insight generation, anomaly explanation, and key finding drafts. All AI outputs reviewed and verified against source data.

KEY FINDING

All analysis conducted using industry-standard tools: Snowflake · SQL · Power BI Desktop · Power BI Service · Claude AI. AI assistance used to accelerate documentation — not to replace analysis.

Known constraints — read before interpreting results

DATA LIMITATIONS

- Pit stop data only available from 2011 onwards
- Fastest lap data is sparse before 2004
- Lap time data begins from 1996 — not 1950
- \N string nulls throughout raw data — handled via NULLIF()
- 'Lotus' refers to two completely unrelated teams across eras
- Grid = 0 entries excluded (pit lane starters)

ANALYTICAL LIMITATIONS

- Points system changed in 2010 — championship gaps not directly comparable pre/post 2010
- Modern era defined as 2000–2024 — some metrics sensitive to this boundary
- Constructor names not standardised across eras
- Constructors with < 150 races excluded from reliability analysis
- No weather or tyre strategy data incorporated

KEY FINDING

Transparency about data limitations is a mark of rigorous analysis. These caveats should be considered when applying findings to real-world strategy decisions.

Three actionable recommendations for the strategy team

01

PRIORITISE QUALIFYING PACE

Data across 75 seasons shows qualifying position is the single strongest predictor of race outcome. 79% of races are won from the top 3 grid slots. Every position gained on Saturday is worth more than any in-race strategy gain on Sunday.

→ Investment priority: one-lap aerodynamic & power unit performance

02

TARGET MERCEDES RELIABILITY BENCHMARK

Mercedes leads modern constructor reliability at 90% finish rate and 3.1% mechanical DNF rate. The gap between best and worst constructors (Mercedes 3.1% vs Toro Rosso 10.4%) directly translates to championship points lost every season.

→ Target: < 5% mechanical DNF rate — benchmark against Mercedes data

03

EXTEND ANALYSIS TO STRATEGY LAYER

This analysis covers macro performance patterns. The next layer — pit stop timing, tyre degradation, and undercut/overcut windows — requires integration of lap time data (1996+) and pit stop records (2011+) currently available in the dataset.

→ Next phase: lap-time telemetry + pit stop strategy optimisation model